

Cloud Computing Concepts & Architecture

1. Cloud Deployment Models

Cloud deployment models define where the infrastructure is hosted, who manages it, and who can access it.

Public Cloud

- **Definition:** Infrastructure owned and managed by third-party cloud providers, shared across the general public via the internet.
- **Key Features:** Pay-as-you-go pricing, high scalability, no hardware maintenance.
- **Examples:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP).

Private Cloud

- **Definition:** Infrastructure dedicated entirely to a single organization, hosted either on-premises or off-premises.
- **Key Features:** High security, data privacy, complete control.
- **Examples:** On-premises Data Centers, AWS Dedicated Hosts.

Hybrid Cloud

- **Definition:** A environment combining **Public Cloud** and **Private Cloud / On-Premises** infrastructure, allowing data and applications to be shared between them.
- **Key Features:** High flexibility, optimized security for sensitive data, cost efficiency.
- **Examples:** Keeping financial databases on-premises while hosting public-facing web applications on AWS.

Community Cloud

- **Definition:** Shared infrastructure used by multiple organizations with common compliance, security, or regulatory requirements.
- **Key Features:** Cost-sharing among specific group members, shared compliance frameworks.
- **Examples:** Government agency clouds, educational research networks.

2. On-Premises vs. Hybrid Cloud

Feature	On-Premises (On-Prem)	Hybrid Cloud
Location	Local company office/data center	Combination of local data center & Public Cloud

Control	100% full physical control	Shared control (Local sensitive data + Cloud apps)
Financial Model	High Upfront Cost (CapEx)	Balanced Cost (CapEx + OpEx)
Best For	Banking, Military, Strict Regulation Data	Growing Businesses, E-commerce, Mixed Workloads

3. Core Architecture Principles

Scalability

The ability of a system to handle increased workloads by adding computing resources (CPU, RAM, or storage).

- **Vertical Scaling (Scale Up):** Adding more power (RAM/CPU) to an existing server.
- **Horizontal Scaling (Scale Out):** Adding more server instances in parallel.

Elasticity

The ability of a cloud system to automatically expand resources during traffic surges and shrink them when demand drops, optimizing costs.

High Availability (HA)

Designing systems to operate continuously 24/7/365 without interruption by eliminating single points of failure.

Single Point of Failure (SPOF)

Any single component in a system that, if it fails, causes the entire system to stop working. Cloud systems use redundancy to eliminate SPOF.

Fault Tolerance

The capacity of a system to continue operating without any downtime even when one or more hardware or software components fail.

Disaster Recovery (DR)

Strategies and processes to restore IT infrastructure and data following a major disaster (e.g., floods, fire, or ransomware attacks).

- **RPO (Recovery Point Objective):** Maximum acceptable data loss duration.

- **RTO (Recovery Time Objective):** Maximum acceptable downtime duration.

4. The 6 Pillars of Cloud Architecture (AWS Well-Architected Framework)

1. **Operational Excellence:** Automating processes, continuously monitoring systems, and constantly improving daily operations.
2. **Security:** Protecting data, identity access (IAM), and systems using encryption and firewalls.
3. **Reliability:** Ensuring systems automatically recover from failure and run consistently as expected.
4. **Performance Efficiency:** Selecting the right resources and maintaining maximum performance under changing demands.
5. **Cost Optimization:** Eliminating unnecessary expenses and using Auto-Scaling to pay only for required resources.
6. **Sustainability:** Minimizing the environmental impact and carbon footprint of cloud workloads through energy efficiency.